

FSF3827 Assignment 3

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Question 1

Part 1

Let C be a minimal cover for the knapsack set K . The corresponding valid inequality (cover cut) is:

$$\sum_{i \in C} x_i \leq |C| - 1$$

Let C' be an extension of set C by a single variable, defined as $C' = C \cup \{j\}$ for some $j \notin C$. The cover cut for C' is given by:

$$\begin{aligned} \sum_{i \in C'} x_i &= \sum_{i \in C} x_i + x_j \leq |C'| - 1 \\ &= \sum_{i \in C} x_i + x_j \leq |C| \\ \implies \sum_{i \in C} x_i &\leq |C| - x_j \end{aligned}$$

Since $x_j \in \{0, 1\}$, we observe that:

$$\sum_{i \in C} x_i \leq |C| - 1 \leq |C| - x_j$$

Thus, the cover cut for C dominates the cover cut for C' because it imposes a more restrictive bound. Consequently, the cover cut for C' is weaker.

Part 2

Lets lift a variable $x_k \notin C'$ into the cover cut for C . The lifted inequality is given by:

$$\sum_{i \in C} x_i + \alpha_{k_1} x_k \leq |C| - 1$$

Where the lifting coefficient α_{k_1} is defined as:

$$\alpha_{k_1} = |C| - 1 - \max \left(\sum_{i \in C} x_i \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right)$$

Similarly, we lift the same variable x_k into the cover cut for C' . The lifted inequality can thus be expressed as:

$$\sum_{i \in C'} x_i + \alpha_{k_2} x_k \leq |C'| - 1 \implies \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j$$

Here the lifting coefficient α_{k_2} is defined as:

$$\begin{aligned}
\alpha_{k_2} &= |C'| - 1 - \max \left(\sum_{i \in C'} x_i | x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C' \cup k \right) \\
&= |C| - \max \left(\sum_{i \in C} x_i + x_j | x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup j \cup k \right) \\
&\geq |C| - \max x_j - \max \left(\sum_{i \in C} x_i | x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right) \\
&\geq |C| - 1 - \max \left(\sum_{i \in C} x_i | x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right) \\
&\geq \alpha_{k_1}
\end{aligned}$$

Thus we can see that:

$$\begin{aligned}
&\sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j \\
\Rightarrow \sum_{i \in C} x_i + \alpha_{k_1} x_k &\leq \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j \\
\Rightarrow \sum_{i \in C} x_i + \alpha_{k_1} x_k &\leq \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - 1 \leq |C| - x_j
\end{aligned}$$

Therefore, the lifted cover cut for C dominates the lifted cover cut for C' , making the latter weaker.

Question 2

Part 1

Initial LP relaxation solution:

Table 1: Optimization Results (Objective Value: 194.2)

Index (i)	Variable 1 ($j = 1$)	Variable 2 ($j = 2$)
1	0.0	1.0
2	0.0	0.0
3	1.0	0.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.0	1.0
8	0.0	0.0
9	0.0	0.0
10	0.6	0.2
11	0.0	0.0
12	0.0	0.0

A minimal cover found is given by the set $C = \{3, 5, 10\}$ with weights (8, 11, 5) respectively. Adding the corresponding cover cut to the model and re-solving gives:

Table 2: Optimization Results (Objective Value: 194.2)

Index (i)	Variable 1 ($j = 1$)	Variable 2 ($j = 2$)
1	0.6	0.4
2	0.0	0.0
3	1.0	0.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.0	1.0
8	0.0	0.0
9	0.0	0.0
10	0.0	0.8
11	0.0	0.0
12	0.0	0.0

A second minimal cover found is given by the set $C = \{3, 5, 1\}$ with weights $(8, 11, 5)$ respectively. Adding the corresponding cover cut aswell to the model and re-solving gives:

Table 3: Optimization Results (Objective Value: 194.2)

Index (i)	Variable 1 ($j = 1$)	Variable 2 ($j = 2$)
1	1.0	0.0
2	0.0	0.0
3	0.0	1.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.6	0.4
8	0.0	0.0
9	0.0	0.0
10	0.0	0.8
11	0.0	0.0
12	0.0	0.0

Which seems to not improve the objective value... The cuts seems ineffective in this case.

Part 2

Introducing sequential lifting for the first minimal cover $C = \{3, 5, 10\}$, we obtain the following optimization results: Doesnt seem to help either... Are our minimal covers just bad?

Question 3

We know that:

$$\begin{aligned}
\sum_{i=1}^n a_i x_i + \sum_{g_i < 0} g_i y_i &\leq \sum_{i=1}^n a_i x_i + \sum_{i=1}^p g_i y_i \leq b \\
\sum_{i=1}^n ([a_i] + f_i) x_i + \sum_{g_i < 0} g_i y_i &\leq [b] + f_0 \\
\Rightarrow \sum_{i=1}^n f_i x_i + \sum_{g_i < 0} g_i y_i - f_0 &\leq [b] - \sum_{i=1}^n [a_i] x_i
\end{aligned}$$

Here we can clearly see that the RHS is an integer:

$$\sum_{i=1}^n f_i x_i + \sum_{g_i < 0} g_i y_i - f_0 \leq K$$

Thus let it be denoted as K and let's see how it behaves for cases when $K \leq -1$ and $K \geq 0$.
Case $K \leq -1$:

$$\begin{aligned} \sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i &\leq \sum_{i=1}^n f_i x_i + \sum_{g_i < 0} g_i y_i \leq K + f_0 \leq K(1 - f_0) \\ \implies \frac{1}{1 - f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) &\leq K \end{aligned}$$

This is because $(f_i - f_0)^+ \leq f_i$, and also when $K \leq -1$ then $K + f_0 \leq K(1 - f_0)$.
Case $K \geq 0$:

$$\begin{aligned} \sum_{i=1}^n f_i x_i + \sum_{g_i < 0} g_i y_i - f_0 &\leq K \\ \implies \sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i &\leq K + f_0 \end{aligned}$$

Now let's split up the LHS based on the set $S = \{i | f_i > f_0\}$ and start with the subcase where $\sum_{i \in S} x_i \geq K + 1$:

$$\begin{aligned} \sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i &= \sum_{i \in S} (f_i - f_0) x_i + \sum_{g_i < 0} g_i y_i \leq \sum_{i \in S} f_i x_i + \sum_{g_i < 0} g_i y_i \\ \implies \sum_{i \in S} f_i x_i + \sum_{g_i < 0} g_i y_i &\leq K + f_0 - f_0 \left(\sum_{i \in S} x_i \right) = K + f_0 \left(1 - \sum_{i \in S} x_i \right) \\ \implies \sum_{i \in S} f_i x_i + \sum_{g_i < 0} g_i y_i &\leq K + f_0 \left(1 - \sum_{i \in S} x_i \right) \leq K + f_0 (1 - (K + 1)) = K(1 - f_0) \\ \implies \frac{1}{1 - f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) &\leq K \end{aligned}$$

For the other subcase where $\sum_{i \in S} x_i \leq K$, we have:

$$\begin{aligned} \sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i &= \underbrace{\sum_{i \in S} (f_i - f_0) x_i + \sum_{g_i < 0} g_i y_i}_{f_i - f_0 < 1 - f_0} < \sum_{i \in S} (1 - f_0) x_i + \sum_{g_i < 0} g_i y_i \\ \implies \frac{1}{1 - f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) &< \sum_{i \in S} x_i + \underbrace{\frac{1}{1 - f_0} \sum_{g_i < 0} g_i y_i}_{\text{Negative}} \leq \sum_{i \in S} x_i \leq K \\ \implies \frac{1}{1 - f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) &< K \end{aligned}$$

Thus we can conclude that the LHS:

$$\frac{1}{1-f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) \leq K \quad \forall K \in \mathbb{Z}$$

And thus substituting back for K gives us the final valid inequality:

$$\begin{aligned} &\Rightarrow \frac{1}{1-f_0} \left(\sum_{i=1}^n (f_i - f_0)^+ x_i + \sum_{g_i < 0} g_i y_i \right) \leq \lfloor b \rfloor - \sum_{i=1}^n \lfloor a_i \rfloor x_i \\ &\Rightarrow \sum_{i=1}^n \left(\lfloor a_i \rfloor + \frac{(f_i - f_0)^+}{1-f_0} x_i \right) + \sum_{g_i < 0} \frac{g_i}{1-f_0} y_i \leq \lfloor b \rfloor \end{aligned}$$

Question 4

Question 5